



PRODUCT REQUIREMENTS DOCUMENT

# ARISE ICT HUB

## Candidate Registration & Certificate Verification Portal

A public web portal for ARISE ICT HUB, Uyo. Prospective students enrol online and receive an auto-generated registration number; graduates verify their record and download an official PDF certificate using two factors. An allowlisted admin console manages candidates and issues certificates — with no privileged key ever reaching the browser.

| PRODUCT        | VERSION       | STATUS             |
|----------------|---------------|--------------------|
| arise-ict-hub  | 0.1.0         | Built — pre-launch |
| DOCUMENT DATE  | OWNER         | PLATFORM           |
| 13 August 2026 | ARISE ICT HUB | Web (responsive)   |

# 1. Overview

---

## 1.1 Problem

ARISE ICT HUB enrolls students onto ICT programmes and awards a certificate on completion. Both processes have been manual: registration on paper, and certificates handed out physically with no way for a student — or a prospective employer — to confirm afterwards that a record is genuine. Paper records are slow to search, easy to lose, and impossible to verify remotely.

## 1.2 Solution

A single responsive web portal that does five things:

- **markets** the hub and its programmes (landing, about and success-story pages);
- takes **student registrations online** and issues a registration number automatically;
- lets a student **verify their record** and download their certificate using their registration number plus the phone number they registered with;
- gives staff an **admin console** to review candidates, add historic (paper-registered) students, and issue or revoke certificates;
- collects and publishes **alumni success stories**, with submissions approved by staff before they appear publicly.

## 1.3 Goals

- **Digitise enrolment** — replace the paper form with a validated online one.
- **Make certificates verifiable** — any holder can prove their record from any device.
- **Protect candidate data** — personal records must never be publicly enumerable.
- **Keep admin work small** — one upload per student issues a certificate.
- **Preserve history** — previously registered students can be brought into the system.
- **Grow a community** — publish alumni experiences that show the hub's real outcomes.

## 1.4 Non-goals (this release)

- Payments, invoicing, or fee collection.
- Course delivery, lessons, assignments, or grading (no learning platform).
- Student self-service login and accounts — verification is credential-based, not session-based.
- Automated certificate generation or templating — the PDF is produced outside the system and uploaded.
- Bulk/CSV import of historic students (added one at a time via the admin console).
- Email or SMS notifications.

## 1.5 Users

| USER                                       | NEEDS FROM THE PRODUCT                                                                                                                        |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prospective student</b>                 | Understand the programmes, then register from a phone in a few minutes and receive a registration number to keep.                             |
| <b>Enrolled student / graduate</b>         | Check whether their certificate has been issued and download the official PDF without creating an account.                                    |
| <b>Administrator</b> (allowlisted staff)   | See who has registered, add students who enrolled on paper, upload each certificate, revoke one issued in error, and moderate alumni stories. |
| <b>Third party</b> (employer, institution) | Confirm a certificate is genuine — possible only with the holder's own two credentials, by design.                                            |

## 2. Scope & Site Map

The product is a responsive web application with a public site and a separate staff console. Public pages share a marketing header and footer; the console renders outside that chrome with its own layout and navigation.

| PAGE               | ACCESS     | PURPOSE                                                                                           |
|--------------------|------------|---------------------------------------------------------------------------------------------------|
| Home               | Public     | Landing page — hero, statistics, programmes, why-choose-us, how it works, testimonials, sponsors. |
| About              | Public     | Mission, story, core values, offerings, strategic goals, call to action.                          |
| Register           | Public     | New student registration form with passport photo upload.                                         |
| Verify Certificate | Public     | Certificate inquiry — two-factor lookup, student profile, certificate download.                   |
| Success Stories    | Public     | Alumni testimonials, curated features and approved student submissions.                           |
| Share a Story      | Public     | Form for alumni to submit their ARISE experience for review.                                      |
| Admin Console      | Admin only | Console: dashboard, students, certificates, stories, settings.                                    |

### 2.1 In scope

- Public marketing pages with the hub's real programme and sponsor content.
- Online registration with server-side validation and server-assigned registration numbers.
- Passport photo upload (JPEG/PNG, maximum 5 MB).
- Two-factor verification: registration number + registered phone number.
- Certificate download via a short-lived signed link (five minutes).
- Admin console with dashboard statistics, searchable/filterable student list, profile view, certificate issue and revoke, manual student creation, and story moderation.
- Alumni success stories with staff approval before publication.
- Full mobile responsiveness including a bottom tab bar and bottom-sheet dialogs on the console.

### 2.2 Out of scope

- Everything listed under non-goals (§1.4).
- Multi-role admin permissions — every allowlisted admin has identical rights.
- Audit logging of admin actions beyond the derived activity feed on the dashboard.
- Editing a candidate's details after creation.

## 3. Functional Requirements

### 3.1 Public website

| ID   | REQUIREMENT                                                                                                                                               | PRIORITY |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| PS-1 | The landing page presents the hub's hero message, headline statistics, programme cards, differentiators, a three-step process, testimonials and sponsors. | Must     |
| PS-2 | The header links to Home, About, Programs and Certificate; the Programs link scrolls to the programmes section on the landing page.                       | Must     |
| PS-3 | On page change the view scrolls to the top, or smoothly to the anchored section when the address carries a section target.                                | Must     |
| PS-4 | The about page shows the hub's mission, story, core values, offerings and strategic goals.                                                                | Should   |
| PS-5 | Every page is fully usable on a phone at 360 px width.                                                                                                    | Must     |

### 3.2 Student registration

| ID    | REQUIREMENT                                                                                                                                                                                                                                    | PRIORITY |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| RG-1  | The form collects: full name, phone, email (optional), date of birth, gender, marital status, state of origin, LGA, address, occupation, religion, last institution, next of kin name and phone, course, class schedule, and a passport photo. | Must     |
| RG-2  | Full name, phone, occupation, religion and passport photo are required. All other fields are optional.                                                                                                                                         | Must     |
| RG-3  | Course is chosen from a fixed list: Web Development, UI/UX Design, Graphic Design, Data Analytics, Basic Computer Education.                                                                                                                   | Must     |
| RG-4  | Class schedule is chosen from: 9am–11am, 11am–1pm, 2pm–4pm.                                                                                                                                                                                    | Must     |
| RG-5  | The passport photo must be JPEG or PNG and no larger than 5 MB; the applicant sees a thumbnail preview and can change or remove it before submitting.                                                                                          | Must     |
| RG-6  | The registration number is generated by the server. The applicant never supplies or influences it.                                                                                                                                             | Must     |
| RG-7  | Registration numbers follow <b>ARISE/ICT/{year}/{0000}</b> , sequential and dense within each year.                                                                                                                                            | Must     |
| RG-8  | The registration year is derived server-side from the current date, never supplied by the applicant.                                                                                                                                           | Must     |
| RG-9  | Every rule is enforced again on the server; on-screen validation is a convenience only.                                                                                                                                                        | Must     |
| RG-10 | When an email is supplied it must be unique; a duplicate is rejected with a clear message.                                                                                                                                                     | Must     |
| RG-11 | On success the applicant sees a confirmation showing their registration number, with a copy button and guidance to keep it safe alongside their phone number for verification.                                                                 | Must     |
| RG-12 | Server errors are translated into plain-English messages; raw technical errors are never shown.                                                                                                                                                | Must     |

### 3.3 Verification & certificate download

| ID    | REQUIREMENT                                                                                                                                                    | PRIORITY |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| VF-1  | Verification requires <b>both</b> the registration number and the phone number used at registration. Neither alone returns anything.                           | Must     |
| VF-2  | Phone numbers are compared on digits only, so spacing, dashes and <code>+234</code> versus leading <code>0</code> do not block a legitimate match.             | Must     |
| VF-3  | A successful lookup shows: passport photo, full name, age, marital status, state of origin, course, and course status (Graduated or In Progress).              | Must     |
| VF-4  | Age is computed from the date of birth on the server; the date of birth itself is never returned.                                                              | Must     |
| VF-5  | The verification response excludes the internal record id, certificate file reference, email and address.                                                      | Must     |
| VF-6  | The download button appears only when the certificate has been issued; otherwise a pending message explains the record is not ready.                           | Must     |
| VF-7  | Download identity is re-proved server-side: the signing service re-checks the registration number and phone against the record before releasing anything.      | Must     |
| VF-8  | The certificate is delivered as a signed link valid for five minutes, served as an attachment named <code>ARISE-Certificate-{registration number}.pdf</code> . | Must     |
| VF-9  | The download starts without navigating away from the profile page.                                                                                             | Should   |
| VF-10 | A failed lookup returns one generic message and never reveals which of the two credentials was wrong.                                                          | Must     |
| VF-11 | A "check another record" action clears the profile and returns to the inquiry form.                                                                            | Should   |

### 3.4 Alumni success stories

| ID   | REQUIREMENT                                                                                                                                | PRIORITY |
|------|--------------------------------------------------------------------------------------------------------------------------------------------|----------|
| ST-1 | The Success Stories page leads with curated testimonials, followed by approved alumni submissions, newest first.                           | Must     |
| ST-2 | Curated stories show a photo; alumni submissions without a photo show an initials chip instead.                                            | Should   |
| ST-3 | Alumni can submit a story with their full name, the programme they took, their phone number and a written experience (40–1200 characters). | Must     |
| ST-4 | A submission is created as <b>Pending</b> ; it never appears publicly until a staff member approves it.                                    | Must     |
| ST-5 | The submitter's phone number is private: shown only to staff, never on the public page.                                                    | Must     |
| ST-6 | On success the submitter sees a confirmation that their story is under review.                                                             | Must     |

### 3.5 Admin console

#### Access control

| ID   | REQUIREMENT                                                                                                                               | PRIORITY |
|------|-------------------------------------------------------------------------------------------------------------------------------------------|----------|
| AD-1 | Admins sign in with email and password through the platform's authentication service.                                                     | Must     |
| AD-2 | Authentication alone is not authorisation. Admin rights come from an account allowlist, checked by the server on every privileged action. | Must     |
| AD-3 | A signed-in user who is not on the allowlist sees a clear "not authorised" screen and can reach no data.                                  | Must     |
| AD-4 | The allowlist itself is unreadable from the browser.                                                                                      | Must     |
| AD-5 | Sign-out is available from the settings tab.                                                                                              | Must     |

#### Dashboard

| ID   | REQUIREMENT                                                                                                                          | PRIORITY |
|------|--------------------------------------------------------------------------------------------------------------------------------------|----------|
| AD-6 | A welcome card greets the admin by time of day and summarises how many certificates are awaiting issue.                              | Should   |
| AD-7 | Statistic tiles show total students, certificates issued, awaiting issue, and registrations this year.                               | Must     |
| AD-8 | A recent-activity feed lists the latest registrations and issues with relative timestamps, and a notification bell flags new events. | Should   |

#### Student management

| ID    | REQUIREMENT                                                                                                               | PRIORITY |
|-------|---------------------------------------------------------------------------------------------------------------------------|----------|
| AD-9  | The students tab lists every candidate with passport thumbnail, name, registration number, course, and issue status.      | Must     |
| AD-10 | The list is searchable by name, registration number, email and phone, and filterable by status (all / issued / awaiting). | Must     |
| AD-11 | The list is paginated, newest registration first.                                                                         | Must     |
| AD-12 | A refresh control reloads the list without a full page reload.                                                            | Should   |
| AD-13 | Opening a student shows their full profile: photo, all captured fields, and certificate state.                            | Must     |

## Certificate operations

| ID    | REQUIREMENT                                                                                                                                    | PRIORITY |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| AD-14 | The admin uploads exactly one certificate per student — a PDF of no more than 5 MB — from the student's profile, by browsing or drag-and-drop. | Must     |
| AD-15 | Certificates are stored privately; re-uploading replaces the existing file.                                                                    | Must     |
| AD-16 | A student is marked "issued" only after the server confirms the file is actually present. A failed upload can never flip the status.           | Must     |
| AD-17 | Issuing sets the verified flag and the issue date, which unlocks the student's download.                                                       | Must     |
| AD-18 | An admin can revoke a certificate: the record returns to awaiting and the stored file is deleted.                                              | Must     |
| AD-19 | An admin can preview an issued certificate through a short-lived signed link.                                                                  | Should   |
| AD-20 | Every operation reports success or failure through an on-screen message.                                                                       | Must     |

## Add student (historic / paper registrations)

| ID    | REQUIREMENT                                                                                                                              | PRIORITY |
|-------|------------------------------------------------------------------------------------------------------------------------------------------|----------|
| AD-21 | An admin can create a candidate record directly, for students who originally registered on paper.                                        | Must     |
| AD-22 | The form collects the same fields as public registration, plus a registration year so past cohorts are numbered correctly.               | Must     |
| AD-23 | Phone and passport photo are mandatory — without them the student could never verify or be identified.                                   | Must     |
| AD-24 | A new <b>ARISE/ICT</b> number is assigned from the same server-side counter as public registration. Old paper numbers are not preserved. | Must     |
| AD-25 | The registration year must be between 2000 and next year inclusive.                                                                      | Must     |
| AD-26 | Creation is permitted only for allowlisted admins, enforced on the server, not in the browser.                                           | Must     |
| AD-27 | On success the list refreshes and the new student's profile opens immediately, so their certificate can be uploaded in the same sitting. | Should   |
| AD-28 | The confirmation message states the new registration number.                                                                             | Must     |

## Story moderation

| ID    | REQUIREMENT                                                                                                                                   | PRIORITY |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------|
| AD-29 | The stories tab lists submissions with author, programme, experience, private phone and status (pending / approved / rejected), newest first. | Must     |
| AD-30 | The queue defaults to Pending — the actionable work — with filters for Approved, Rejected and All.                                            | Should   |
| AD-31 | Approving a story makes it appear on the public Success Stories page; rejecting keeps it hidden.                                              | Must     |
| AD-32 | A count badge shows how many stories are awaiting review.                                                                                     | Should   |



Console experience

| ID    | REQUIREMENT                                                                            | PRIORITY |
|-------|----------------------------------------------------------------------------------------|----------|
| AD-33 | On desktop the console uses a left sidebar; on mobile it switches to a bottom tab bar. | Must     |
| AD-34 | Dialogs become bottom sheets on small screens and scroll within the viewport.          | Must     |
| AD-35 | Add Student is reachable from a floating action button on every screen size.           | Should   |
| AD-36 | Dialogs close on Escape and on clicking the backdrop.                                  | Should   |

## 4. Key User Flows

---

### 4.1 Register as a new student

1. The applicant opens the Register page from the site header or a call to action.
2. They complete the form and attach a passport photo, seeing a preview before submitting.
3. The photo uploads to public photo storage under a random file name.
4. The registration process re-validates every field, rejects a duplicate email, derives the current year, and assigns the next number in sequence.
5. The applicant sees a confirmation screen with their registration number (copyable) and is told to keep it with the phone number they used.

### 4.2 Verify a record and download a certificate

1. The student opens the Verify page and enters their registration number and phone number.
2. The verification process matches both factors, comparing the phone on digits only, and returns a narrow profile — or nothing.
3. The profile renders with the photo, personal details and status. If no certificate has been issued, a pending message shows instead of a download button.
4. If issued, the student presses download. The signing service re-checks both credentials, confirms the certificate exists, and mints a five-minute signed link.
5. The browser saves the PDF as an attachment without leaving the page.

### 4.3 Issue a certificate

1. The admin signs in at the console; the server confirms allowlist membership.
2. They filter the students list to *Awaiting* and open a candidate.
3. They attach the candidate's PDF ( $\leq 5$  MB) and confirm the upload.
4. The file is written to private certificate storage.
5. The issue process verifies the file is present, then sets the certificate reference, verified flag and issue date. If the file is missing the call fails and the status is unchanged.
6. The student's download unlocks immediately on the Verify page.

#### 4.4 Add a paper-registered student

1. The admin presses the floating + button on the students tab.
2. They complete the add-student dialog, including the original registration year, phone number and passport photo.
3. The photo uploads, then the admin-only creation process re-checks the allowlist, validates every field, and assigns a fresh number for that year.
4. The list refreshes, the new profile opens, and the certificate can be uploaded straight away.

#### 4.5 Share and moderate a story

1. An alumnus opens Share a Story, enters their name, programme, phone and experience, and submits.
2. The submission is stored as *Pending*; a confirmation explains it will appear once reviewed.
3. The admin opens the Stories tab, sees the Pending queue, and approves or rejects the submission.
4. Approved stories appear on the public Success Stories page; the submitter's phone stays hidden.

## 5. Data & Privacy

### 5.1 Student record

One record per student, created either by public registration or by an admin.

| ITEM                          | CAPTURED AT     | NOTES                                                                      |
|-------------------------------|-----------------|----------------------------------------------------------------------------|
| Full name                     | Required        | Minimum two characters.                                                    |
| Phone number                  | Required        | The second verification factor; a record without it can never be verified. |
| Email                         | Optional        | Must be unique when supplied.                                              |
| Registration number           | Server-assigned | Unique, format <b>ARISE/ICT/{year}/{0000}</b> .                            |
| Registration year             | Server-derived  | Or chosen by an admin for a historic cohort.                               |
| Passport photo                | Required        | JPEG/PNG, $\leq 5$ MB.                                                     |
| Occupation, religion          | Required        | Enforced at the database level.                                            |
| Date of birth                 | Optional        | Used to compute age on the profile; never returned directly.               |
| State of origin, LGA, address | Optional        | State of origin appears on the public profile; address does not.           |
| Gender, marital status        | Optional        | Marital status appears on the public profile.                              |
| Course, class schedule        | Optional        | Course appears on the public profile.                                      |
| Last institution              | Optional        | Admin view only.                                                           |
| Next of kin name and phone    | Optional        | Admin view only.                                                           |
| Certificate state             | Admin-managed   | Hidden until issued; includes reference, verified flag and issue date.     |

### 5.2 Supporting records

- **Registration counters** — one counter per year holding the last sequence issued, keeping numbers dense and collision-free.
- **Admin allowlist** — the set of email addresses permitted to administer the portal. Unreadable from the browser; an admin is added by inserting their address after their sign-in account exists.
- **Story submissions** — author, programme, experience, private phone, and a moderation status (pending / approved / rejected).

## 5.3 Storage

| STORE                  | VISIBILITY | CONTENTS AND RULES                                                                                                                                                                       |
|------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Passport photos</b> | Public     | Photos stored under random file names so profiles can render. JPEG or PNG, $\leq 5$ MB.                                                                                                  |
| <b>Certificates</b>    | Private    | One PDF per student. No public access of any kind. Read and written only by allowlisted admins or the signing service; students receive five-minute signed links. PDF only, $\leq 5$ MB. |

## 6. Security & Privacy Requirements

### GOVERNING PRINCIPLE

**Never trust the client.** The browser holds only public configuration and, for staff, a signed-in session. Every rule that matters — who may read, who may write, what a valid record looks like, and what a registration number may be — is enforced by the server.

| ID    | REQUIREMENT                                                                                                                                                       |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SC-1  | There is no broad public read of candidate records. Records are hidden by default; the public reaches data only through two vetted operations.                    |
| SC-2  | Public reads require both the registration number and the phone number. Records cannot be enumerated with a registration number alone.                            |
| SC-3  | The public profile returns a deliberately narrow field set. Internal id, certificate reference, email, address and date of birth are excluded.                    |
| SC-4  | Writes are only possible through vetted operations. No direct edit or insert of records is granted to visitors or signed-in users.                                |
| SC-5  | The certificate store is private with no anonymous access at all.                                                                                                 |
| SC-6  | Certificate links expire after five minutes and are minted per request.                                                                                           |
| SC-7  | The privileged service key exists only as a server-side secret in the signing service. It is never bundled, exposed, or sent to a browser under any circumstance. |
| SC-8  | The signing service independently re-verifies both credentials before releasing a link — it does not trust that the caller already passed verification.           |
| SC-9  | The signing service refuses to sign when a certificate has not been issued.                                                                                       |
| SC-10 | Admin authority comes from an allowlist evaluated on the server for every privileged action. Modified client code cannot grant it.                                |
| SC-11 | The allowlist is unreadable from the browser.                                                                                                                     |
| SC-12 | Admin read access to candidates is scoped to allowlisted admins; anonymous visitors are never granted it.                                                         |
| SC-13 | Registration numbers are generated exclusively server-side by an internal routine the public cannot call directly.                                                |
| SC-14 | Registration years are derived from the server clock for public registration; an admin-supplied year is range-checked.                                            |
| SC-15 | File type and size limits are enforced for both uploads: images (JPEG/PNG) and certificates (PDF), each $\leq 5$ MB.                                              |
| SC-16 | Certificate issue is transactionally safe: the status cannot be set unless the file is confirmed present.                                                         |
| SC-17 | Error messages are generic. Failed verification never discloses which credential was wrong, and internal errors are never surfaced verbatim.                      |

#### ACCEPTED TRADE-OFFS

**Passport photos are publicly readable.** They sit under unguessable random file names, but anyone holding the exact link can view the image. This is accepted so profiles render without a signing round-trip; the link is only disclosed after a successful two-factor lookup.

**Verification is credential-based, not rate-limited in the application.** Guessing requires a valid registration number and its matching phone number together. Adding request throttling at the edge is a recommended follow-up.

**Every admin is equal.** There are no tiers; anyone on the allowlist can add students and issue or revoke certificates.

## 7. Non-Functional Requirements

| AREA                   | REQUIREMENT                                                                                                                                                             |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsiveness</b>  | Every page works from 360 px upward. The console switches to a bottom tab bar and bottom-sheet dialogs on small screens; tables reflow into cards.                      |
| <b>Performance</b>     | The site loads quickly on typical mobile connections; heavier pages load on demand. Uploads are capped at 5 MB. Verification is a single indexed lookup.                |
| <b>Accessibility</b>   | Labelled form controls, dialog roles on dialogs, accessible names on icon-only buttons, alerts announced to assistive technology, Escape closes dialogs, visible focus. |
| <b>Usability</b>       | Loading states on every asynchronous action, plain-English errors, and a confirmation message after each admin operation.                                               |
| <b>Reliability</b>     | Certificate status cannot desynchronise from stored files. A failed upload leaves the record untouched.                                                                 |
| <b>Maintainability</b> | Validation rules and option lists live in one shared module used by both the public form and the admin form.                                                            |
| <b>Browser support</b> | Current versions of Chrome, Edge, Firefox and Safari, on desktop and mobile.                                                                                            |
| <b>Privacy</b>         | Only what a profile needs is returned publicly. Contact details, address and next of kin remain admin-only. Story submitters' phone numbers are never public.           |



## 8. Status & Future Work

### 8.1 Delivered

| CAPABILITY                      | STATUS | NOTE                                                      |
|---------------------------------|--------|-----------------------------------------------------------|
| Landing and about pages         | Built  | Real programme, sponsor and testimonial content.          |
| Student registration            | Built  | Server validation, server-assigned numbers, photo upload. |
| Two-factor verification         | Built  | Narrow profile, computed age, status badge.               |
| Certificate download            | Built  | Five-minute signed links from the signing service.        |
| Admin authentication and gate   | Built  | Account allowlist, enforced server-side.                  |
| Dashboard, list, search, filter | Built  | Statistics, activity feed, pagination.                    |
| Issue and revoke certificates   | Built  | File-presence check before status change.                 |
| Add student (historic)          | Built  | Admin-only creation with a chosen registration year.      |
| Alumni stories and moderation   | Built  | Public submission, staff approval, private phone.         |
| Mobile console experience       | Built  | Tab bar, bottom sheets, floating action button.           |

### 8.2 Outstanding before launch

- Apply the current schema — including the admin creation procedures — to the production database.
- Add the live administrator addresses to the allowlist.
- Run the end-to-end smoke test against production.

### 8.3 Candidate future work

| ITEM                                      | VALUE  | RATIONALE                                                |
|-------------------------------------------|--------|----------------------------------------------------------|
| Rate limiting on verification and signing | High   | Removes the main brute-force surface that remains.       |
| Edit an existing candidate                | High   | Today a typo cannot be corrected from the console.       |
| Bulk import of historic students          | Medium | One-at-a-time entry does not scale to a large backlog.   |
| Admin action audit log                    | Medium | Records who issued or revoked what, and when.            |
| Email or SMS on issue                     | Medium | Students currently have to check back themselves.        |
| Certificate generation from a template    | Medium | Removes the manual step of producing each PDF.           |
| Public QR verification for employers      | Low    | Third-party checking without the student's phone number. |
| Private passport storage                  | Low    | Closes the accepted trade-off in §6.                     |

## 8.4 Success measures

- The share of new enrolments completed online rather than on paper.
- Successful verifications as a proportion of attempts (a low rate suggests students are losing their credentials).
- Median time from programme completion to certificate issued.
- Certificate downloads per issued certificate.
- Registration form completion rate — starts versus submissions.
- Story submissions approved and published per quarter.